

Predicting Contract Renewal in Public Digital Procurement

A Survival-Analysis Approach on BOAMP Notices for Gigalis
Internship Report — Quick Orientation to What Was Done and Where

Internship Report — Gigalis (Pays de la Loire)

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Abstract

This report is the **quick-reading guide** to the internship: it is meant to let a reader scan the project rapidly and understand *what was done*, *why it matters*, and *where each workstream sits* in the overall pipeline. The detailed implementation, audit logic, intermediate diagnostics, and methodological justifications are intentionally left to the separate **technical report** (`phase1_technical_report.pdf`).

The internship work concerns predictive analysis of public digital procurement for **Gigalis**, a central purchasing body in Pays de la Loire. The goal is to move from a descriptive view (“what happened?”) towards a predictive one (“when will a contract be re-tendered?”). Because BOAMP contains *no explicit renewal field*, the renewal relationship must be reconstructed from observable signals. We build a proxy renewal event using Sentence-Transformer text embeddings combined with CPV, temporal, and buyer signals, and use it as the event variable for survival analysis. On 1,100 eligible source contracts the linking method recovers a plausible renewal for 697 of them (**63.4%**), against an 11.3% lexical baseline. Survival modeling gives a global Kaplan–Meier median survival of **48.1 months**, a Cox concordance of **0.6524**, and a best-fitting log-normal parametric model. The fitted model is then translated into 12- and 24-month renewal-risk indicators at contract, buyer, and segment level.

1 How to Read This Report

This document is a **high-level navigation guide**. Each section states *what* was done, *what the key result is*, and *where* the outputs live. When you need the exact field logic, preprocessing rules, linking formula, diagnostics, or full model specification, go directly to the **technical report**.

Pipeline at a glance

Table 1: Component status and key outputs. Status: **Done**, **Partial**, **Not started**.

Component	Status	Key result	Where
Data cleaning & EDA	Done	3,181 notices → 1,933 source contracts	data/processed/boamp_full_clean.csv
Renewal linking (event reconstruction)	Done	697/1,100 eligible linked (63.4%)	boamp_renewal_linking_quality/
Survival dataset construction	Done	1,100 rows × 36 cols	data/processed/boamp_phase2_survival.csv
Kaplan–Meier & log-rank	Done	Median 48.1 mo; duration significant	notebooks/02_survival_modeling_boamp.ipynb
Cox PH model	Done	C-index 0.6524	same notebook + reports/tables/survival/
Parametric AFT models	Done	Log-normal best (AIC 7,119.8)	same notebook
Sensitivity analysis	Done	Ranking stable across 3 definitions	reports/tables/survival/sensitivity_comparison.csv
Robustness checks	Done	Placebo, bias check, threshold sweep	validation_robustness/
Operational risk indicators	Done	12/24-mo risk at contract/buyer/segment	reports/tables/survival/top20_renewal_risk.csv
Technology taxonomy	Done	Rule-based (CPV + keywords), 11 categories	data/processed/taxonomy.csv
Manual event validation	Partial	Sample drawn; review not done	event_validation/
Trained NLP classifier	Not done	Taxonomy is rule-based only	—
Change-point detection (Phase 4)	Not done	Scoped but not implemented	—

2 Context and Problem

2.1 Why this project

Gigalis operates a central purchasing body for digital services in Pays de la Loire. To anticipate re-tendering workloads, it wants to predict *when* digital contracts come up for renewal. The core obstacle: BOAMP publishes calls for tenders and awards, but *no notice says “this renews contract X”*. The renewal link must be inferred from text, classification, timing, and buyer identity.

2.2 Scope

- **Sectors:** CPV divisions 48 (software), 72 (IT services), 32 (telecom), 35 (security).
- **Territory:** departments 44, 49, 53, 72, 85 (Pays de la Loire).
- **Period:** 2015–2024. Study end date: 2024-12-31.
- **Modeling path:** BOAMP only. DECP is documented but not linked (no shared key between the two sources).

3 Data

3.1 Corpus numbers

- 3,181 BOAMP notices downloaded (2015–2024).
- 1,933 are APPEL_OFFRE (source contracts); 1,081 are ATTRIBUTION.

- 1,100 are *eligible*: their expected renewal window falls before 2024-12-31.
- 833 are structurally censored upfront (too recent to observe a renewal).

DECP (3,039 digital contracts, same scope) is far richer (100% SIRET, 99.4% durations) but cannot be merged to BOAMP at contract level. It remains a future enrichment path.

3.2 Key data quality issues (modeling consequences)

Table 2: Key BOAMP data-quality findings and their modeling consequence.

Finding	Value	Consequence
Buyer SIRET coverage	~9% of corpus	Grouping uses normalised names; same entity may fragment into several keys.
Amount fill rate (APPEL_OFFRE)	~21%	Amount excluded from linking signals.
Declared duration granularity	69% at 12/24/36/48 mo	Duration used as a covariate, not as survival target.
Imputed durations	25% of eligible cohort	Flagged with <code>dur_was_imputed</code> ; imputation is 48 months.
No explicit renewal label	structural	<code>event</code> is a proxy reconstructed by the linking algorithm.

Full field-level cleaning rules and missingness tables are in the technical report (§2) and `reports/data_quality_report/`.

3.3 Taxonomy

Each contract is tagged with one of eleven technology categories (Software & Applications, IT Services & Consulting, Telecom & Networks, Cybersecurity, Cloud & Infrastructure, Data & AI, etc.) via a **deterministic rule-based mapping** from CPV codes and description keywords. This is *not* a trained classifier — the planned NLP classifier was not implemented. Output: `data/processed/taxonomy.csv`.

4 What Was Built

4.1 Phase 1 — Renewal-event reconstruction

Each eligible source contract is matched against later BOAMP notices from the same buyer using a four-signal composite score: semantic text embeddings (Sentence-Transformer, model `paraphrase-multilingual-MiniLM-L12-v2`), CPV hierarchy match, temporal proximity, and buyer-identity agreement (weights 0.40 / 0.25 / 0.20 / 0.15). The best candidate above the hard filter threshold is selected as the renewal successor. See technical report §3.2 for the full formula and weight derivation.

Table 3: Renewal-linking outcome.

Quantity	Value	
Source contracts (<code>APPEL_OFFRE</code>)	1,933	
Eligible (renewal window observable)	1,100	
Linked renewals (<code>event</code> = 1)	697	(63.4%)
Right-censored (<code>event</code> = 0)	403	(36.6%)
Lexical baseline linking rate	11.3%	

Linking rate by category (Table 4): Software & Applications and IT Services & Consulting renew most (71%, 68%); Data & AI and IT Hardware least (41%, 48%). Small categories ($n < 10$) are underpowered.

Table 4: Event rate by technology category (1,100 eligible contracts).

Category	n	events	rate %
IT Services & Consulting	312	211	67.6
Software & Applications	209	148	70.8
Telecom & Networks	204	120	58.8
Cybersecurity	121	76	62.8
Unknown	108	58	53.7
Digital Workplace & Collaboration	45	29	64.4
Data & AI	34	14	41.2
IT Maintenance & Support	22	14	63.6
IT Hardware & Equipment	21	10	47.6
Cloud & Infrastructure	19	13	68.4
GIS & Mapping	5	4	80.0

Link quality. Of the 697 accepted links: 135 are single-candidate (no competing match); of the remaining 562 multi-candidate links, 273 (49%) have a score margin below 0.05, flagging an ambiguous winner. Only 53 links (7.6%) satisfy the strict high-confidence criterion (score ≥ 0.80 and margin ≥ 0.05). This low high-confidence rate motivates the manual validation step (not yet done; see Section 5).

Main unlinked failure mode. 339 of 403 unlinked contracts (84%) have no temporal partner in the search window — not a scoring failure, but a true absence of a candidate notice in the observable period.

Outputs: `boamp_renewal_linking_quality/outputs/` (`boamp_renewal_links.csv`, `boamp_renewal_candidates.csv`, `boamp_linking_stats.csv`).

4.2 Phase 2 — Survival analysis

The survival dataset (`data/processed/boamp_phase2_survival.csv`) has one row per eligible contract with observed/censored duration and covariates (`declared_duration_months`, `dur_was_imputed`, `start_year`, category, procedure type). Three model families were fitted in `notebooks/02_survival_modeling_boamp.ipynb`.

Kaplan–Meier. Global median survival: **48.1 months**, consistent with the prevalence of 48-month framework contracts. Log-rank tests (Table 5): renewal timing is driven by declared duration and imputation status ($p < 0.001$), not by technology category ($p = 0.15$).

Table 5: Log-rank tests for differences in renewal timing.

Stratifying variable	statistic	p	df
Category (multivariate)	6.75	0.150	4
Duration was imputed	24.41	< 0.001	1
Declared-duration group	62.43	< 0.001	2
Procedure type (collapsed)	2.10	0.349	2

Cox proportional-hazards model. In-sample concordance: **0.6524**. Two predictors are significant (Table 6): longer declared duration lowers hazard (HR 0.991 per month), and later start year raises it (HR 1.094 per year). No category or procedure effect.

Table 6: Cox model: significant covariates (multivariate).

Covariate	HR	95% CI	p
declared_duration_months	0.991	[0.987, 0.995]	< 0.001
start_year	1.094	[1.050, 1.140]	< 0.001
dur_was_imputed	0.864	[0.718, 1.040]	0.123
Category / procedure terms	not significant ($p > 0.1$)		

The positive `start_year` HR is partly structural: 2021–2024 contracts are right-censored because a typical 48-month contract would renew after the 2024-12-31 study close.

Parametric AFT models. Log-normal fits best (AIC 7,119.8, concordance 0.672), outperforming log-logistic (7,194.3) and Weibull (7,320.3). Full parametric results: `reports/tables/survival/parametric_aic_comparison.csv`.

Sensitivity analysis. Three event definitions were tested (Table 7). The Cox concordance is stable across all three, confirming that the *ranking* of renewal risk is robust even under the strictest definition.

Table 7: Sensitivity to event definition. “–” = KM median never reached. KM median in months.

Variant	events	rate %	KM median	Cox C
A — baseline (all links)	697	63.4	48.1	0.6524
B — conservative (≥ 0.50)	494	44.9	52.7	0.6436
C — strict H.C. ($S \geq 0.70$, $\Delta \geq 0.05$, multi-cand.)	90	8.2	–	0.6295

Full results: `reports/tables/survival/` (9 CSV files: univariate, multivariate, log-rank, AIC, sensitivity, top-20 risk ranking).

Proportional hazards assumption. Schoenfeld test flags a violation for `declared_duration_months` ($p < 10^{-5}$); all other covariates pass. Stratification or a time interaction for duration is flagged as future work. The log-normal AFT avoids the PH assumption and gives the same conclusions.

4.3 Robustness and validation checks

These are in `validation_robustness/` (untracked — results exist but not merged to main git history yet).

- **Threshold sensitivity (7 scenarios, score 0.35 to 0.90):** Cox concordance stays above 0.59 throughout; directional findings stable. Full table: `validation_robustness/outputs/threshold_sensitivity_summary.csv`.
- **Placebo check:** winning link mean score 0.579 vs. runner-up 0.435 vs. synthetic no-buyer 0.425 — scoring discriminates.
- **Event/censoring bias:** SMD = 0.13 (duration), -0.03 (start year) between event and censored groups — covariates balanced; dominant bias is structural right-censoring of 2022–2024 contracts.
- **Manual validation:** 150-row sample drawn (`event_validation/outputs/manual_validation_sample.csv`), protocol documented (`event_validation/event_validation_protocol.md`). **Record-level review not yet executed.** Precision/recall of the linking step therefore unknown.

4.4 Operational renewal-risk indicators

The log-normal AFT model is applied to all 1,097 scored contracts to produce conditional renewal probabilities at 12 and 24 months ahead.

Table 8: Operational risk summary (1,097 scored contracts).

Metric	Value
Mean 12-month renewal probability	0.190
Mean 24-month renewal probability	0.332
Max 12-month probability (any contract)	0.435
Contracts in <i>medium</i> band (0.25–0.50)	222
Contracts in <i>low</i> band (< 0.25)	875
Expected renewals within 12 months	208
Expected renewals within 24 months	365
Top buyer by expected 12-month volume	Pays de la Loire (18.6)
Top segment by expected 12-month volume	IT Services & Consulting (61.6)

These are prioritisation indicators, not legally verified renewal probabilities. Output: `reports/tables/survival/top20_renewal_risk.csv`.

5 What Was Not Done

- **Manual validation of linked pairs.** The 150-row sample is ready; the review has not been executed. Until done, there are no ground-truth precision / recall estimates for the linking step.
- **Trained NLP classifier.** The technology taxonomy is rule-based (CPV + keywords). The planned learned classifier was not implemented.
- **Change-point / trend detection (Phase 4).** Scoped but not started. No code or outputs exist.
- **DECP semantic linking.** The Sentence-Transformer method was applied to BOAMP only. DECP linking remains at the 8.3% lexical prototype.
- **Live prediction scoring.** The current model evaluates contract age as of the study end date. A production run would need to re-evaluate each contract at a live prediction date.

6 Key Limitations

- **Proxy event.** `event = 1` means a plausible renewal link was found, not a legally certified renewal. Most links are low-to-medium confidence (48.6% low, 24.4% medium, 7.6% high).
- **Weak buyer identification.** Only $\sim 13\%$ of buyers carry a SIRET; the rest are matched by normalised name. Spelling variants (“Ville de Nantes” vs “Nantes Métropole”) create false negatives. The 2024 eForms notices do include SIRET but are all censored upfront.
- **Short-text false positives.** Generic descriptions (“prestations informatiques”) can match spuriously.
- **Structural right-censoring.** Contracts from 2021–2024 are largely censored; the positive `start_year` hazard partly reflects study truncation, not a genuine cohort trend.
- **Modest discrimination.** Concordance ≈ 0.65 — useful for prioritisation, not sufficient for high-stakes individual decisions.

7 Conclusion and Next Steps

The internship delivered a reproducible end-to-end pipeline: BOAMP notice acquisition $\rightarrow$ cleaning $\rightarrow$ renewal-event reconstruction $\rightarrow$ survival modeling $\rightarrow$ operational risk indicators. The main methodological result is lifting the renewal linking rate from 11.3% (lexical baseline) to 63.4% using Sentence-Transformer embeddings. Survival modeling shows renewal timing is driven by declared contract duration (median 48.1 months) rather than technology family, and the log-normal AFT model translates this into actionable 12/24-month renewal-risk rankings.

Immediate next steps (prioritised):

1. **Execute manual validation** of the 150-row sample to get ground-truth precision/recall for the linking step (`event_validation/`).
2. **Re-score contracts at a live prediction date** (current scores use end-of-study ages).
3. **SIRET/SIREN enrichment** to strengthen buyer resolution.
4. **Replace rule-based taxonomy** with a trained text classifier.
5. **Implement change-point detection** (Phase 4).